II.F Performance and Limitations

Lesson Plan. Area of Operation II, Technical Subject Areas. ACS Task code AI.II.F.

[Name], CFI Applicant

Ground lesson. Uses the matching slide deck "II.F Performance and Limitations - Slides" (42 slides).

1. Objective

The student will use the C172S POH performance charts, tables, and data to predict takeoff, climb, cruise, fuel, and landing performance, explain the factors that change performance, compute weight and balance, correct an out-of-limits CG, and confirm the airplane stays within limits for the whole flight.

ACS objective: To determine the applicant understands aircraft performance and limitations, can apply that knowledge, manage associated risks, demonstrate appropriate skills, and provide effective instruction.

2. Elements

ACS code	Element	Slides
AI.II.F.K1	Elements related to performance and limitations by explaining the use of charts, tables, and data to determine performance.	4, 5, 6, 27
AI.II.F.K2	Factors affecting performance, including:	7, 8, 9, 10, 11, 12, 13
AI.II.F.K2a	a. Atmospheric conditions	7, 8
AI.II.F.K2b	b. Pilot technique	9
AI.II.F.K2c	c. Airplane configuration	10
AI.II.F.K2d	d. Airport environment	11, 12
AI.II.F.K2e	e. Loading and weight and balance	13
AI.II.F.K3	Weight and balance terms, including: basic empty weight, maximum gross weight, arm, moment, reference datum, center of gravity (CG) and CG limits, and useful load.	14, 15, 16, 17, 18, 19, 20, 21, 22
AI.II.F.K4	Methods for computing CG.	23, 24
AI.II.F.K5	Aerodynamics.	25, 26
AI.II.F.R1	Use of performance charts, tables, and data.	37
AI.II.F.R2	Airplane limitations.	14, 20, 38
AI.II.F.R3	Possible differences between calculated performance and actual performance.	37
AI.II.F.R4	Exceeding weight limits.	38
AI.II.F.R5	Operating outside of CG limits.	39
AI.II.F.R6	Shifting, adding, and removing weight.	35, 36, 39
AI.II.F.S1	Use the appropriate airplane performance charts, tables, and data.	27, 28, 29, 30, 31, 32, 40
AI.II.F.S2	Compute the weight and balance, correct out-of-center of gravity loading errors and determine if the weight and balance remains within limits during all phases of flight.	33, 34, 35, 36, 40

3. Schedule

Part	Slides	Time
Introduction (attention, motivation, overview)	Slides 1 to 3	5 min

Development: charts and factors	Slides 4 to 13	15 min
Development: TCDS, weight and balance terms and methods	Slides 14 to 26	15 min
Development: C172S worked examples	Slides 27 to 36	20 min
Development: risk management	Slides 37 to 39	5 min
Conclusion and review	Slides 40 to 42	5 min
Total		65 min

4. Equipment

- Slide deck (42 slides)
- Whiteboard and markers
- C172S POH/AFM and the airplane's current weight and balance record
- TCDS 3A12 (printed page 30)
- E6B or electronic flight computer, and a calculator
- Blank weight and balance form and nav log
- Model airplane and a ruler (to show datum, arm, and balance)
- Chart Supplement entry for a local airport

5. Introduction

Attention: Ask: "It is 35 degrees C, you have three friends and full tanks, and the runway is 2,500 feet of grass. Can you go?" Most students say yes. By the end of this lesson they will know exactly how to find out.

Motivation: Overloaded airplanes and bad performance planning cause takeoff and landing accidents. The PHAK says the majority of pilot-caused aircraft accidents occur during the takeoff and landing phase of flight (PHAK 11-12). These skills are also tested on every checkride from private pilot up.

Overview: We will cover where performance data is found, how to read charts, what changes performance, the weight and balance terms, how to compute CG, worked C172S examples straight from the POH, and the risks to manage.

6. Development

Teach each slide in order. Slide numbers match the deck exactly.

Slide 1: Performance and Limitations

ACS: AI.II.F

Show the title slide while you give the attention step from the Introduction. The photos show the same airplane on pavement and on grass. Ask the student which one needs more runway and why. The answer comes later in the lesson.

Source: PHAK Fig 11-16

Slide 2: Objective

ACS: AI.II.F

State the objective in plain words. The student will use the charts, tables, and data in the C172S POH to predict takeoff, climb, cruise, fuel, and landing performance. The student will compute weight and balance, correct a CG that is out of limits, and check that weight and balance stays in limits for the whole flight.

Source: ACS II.F, POH Fig 5-5

Slide 3: Lesson Overview

ACS: AI.II.F

Give a short road map: charts and performance factors, weight and balance terms and methods, worked C172S examples, then risk management and review.

Source: POH Fig 6-7

Slide 4: Where the Numbers Live

ACS: AI.II.F.K1

The performance section of the AFM/POH holds the data for takeoff, climb, range, endurance, descent, and landing. The PHAK says using this data is mandatory for safe and efficient operation (PHAK 11-1).

Show the C172S Section 5 table of contents. Point out each chart the student will use today: takeoff (Figure 5-5), climb (Figure 5-7), cruise (Figure 5-8), range (Figure 5-9), and landing (Figure 5-11).

CG limits are published in the Type Certificate Data Sheet (TCDS) and in the AFM or POH (PHAK 10-3).

Ask the student:

- Which POH section would you open to find the maximum takeoff weight?
- Which section has the takeoff distance chart?

Source: PHAK 11-1, PHAK 10-3, POH 5-1

Slide 5: How the Charts Are Made

ACS: AI.II.F.K1

Performance charts come from test flights conducted in a new aircraft, under normal operating conditions, using average piloting skills, with the aircraft and engine in good working order (PHAK 11-19).

The C172S POH says the data was computed from actual flight tests with the airplane and engine in good condition and approximating average piloting techniques (POH 5-3).

Every chart has a conditions block (flaps, power, runway, wind). If your situation does not match, the number does not match either.

Example: The C172S short field takeoff chart assumes flaps 10 degrees, full throttle before brake release, a paved, level, dry runway, and zero wind (POH Fig 5-5).

Ask the student:

- Is our airplane new? What does that mean for chart numbers?
- What is the first thing you read on any chart?

Source: PHAK 11-19, POH 5-3, PHAK Fig 11-20

Slide 6: Reading Tables and Interpolating

ACS: AI.II.F.K1

Interpolating means using known values on the chart to compute values in between (PHAK 11-20).

The PHAK example: for 2,500 feet pressure altitude at 20 degrees C, average the 2,000 foot and 3,000 foot ground rolls: (1,115 + 1,230) divided by 2 = 1,173 feet (PHAK Fig 11-21).

The C172S POH says conservative distances can be found by reading the chart at the next higher value of weight, altitude, and temperature (POH 5-4). This is faster and gives a safety margin.

Ask the student:

- Field elevation is 1,500 feet. Which row would you use for a conservative answer?

Source: PHAK 11-20, POH 5-4, PHAK Fig 11-21

Slide 7: Definition: Density Altitude

ACS: AI.II.F.K2a

Read the definitions from PHAK 11-3 word for word: density altitude is "the altitude in the standard atmosphere corresponding to a particular value of air density." The PHAK also says "Density altitude is pressure altitude corrected for nonstandard temperature."

Show the density altitude chart (PHAK Fig 11-4). Enter with outside air temperature, go up to the pressure altitude line, then read density altitude on the left.

Source: PHAK 11-3, PHAK Fig 11-4

Slide 8: Density Altitude in Plain Words

ACS: AI.II.F.K2a

As air becomes less dense, it reduces power (the engine takes in less air), thrust (the propeller is less efficient in thin air), and lift (thin air exerts less force on the airfoils) (PHAK 11-2).

Moist air is lighter than dry air, so high humidity raises density altitude and lowers performance. There is no rule of thumb or chart for humidity, but expect a decrease in performance (PHAK 11-5).

Example: On the C172S takeoff chart at 2,550 lb and sea level, total distance over a 50 foot obstacle grows from 1,465 feet at 0 degrees C to 1,945 feet at 40 degrees C. At 8,000 feet and 40 degrees C it is 4,615 feet (POH Fig 5-5).

Ask the student:

- Why does the same airplane need more runway in July than in January?
- Name the three things thin air takes away.

Source: PHAK 11-2, PHAK 11-5, POH Fig 5-5

Slide 9: Pilot Technique

ACS: AI.II.F.K2b

The data assumes average piloting technique (POH 5-3). If skills are below average, the PHAK says to compensate for the chart numbers (PHAK 11-19).

The PHAK says proper takeoff speed is especially important when runway lengths and takeoff distances are critical (PHAK 11-15).

The C172S short field takeoff chart lists lift off at 51 KIAS and 56 KIAS at 50 feet, and the note says to use short field technique as specified in Section 4 (POH Fig 5-5).

Ask the student:

- What happens to takeoff distance if you rotate late and fast?
- Where do you find the technique the chart assumes?

Source: POH 5-3, POH Fig 5-5, PHAK 11-15, PHAK 11-19

Slide 10: Airplane Configuration

ACS: AI.II.F.K2c

Configuration means flaps, gear, and power setting. Lowering the flaps decreases both excess thrust and excess power, which hurts climb (PHAK 11-8).

C172S examples: the short field takeoff chart assumes flaps 10 degrees (POH Fig 5-5). The climb chart assumes flaps up and full throttle (POH Fig 5-7). The short field landing chart assumes flaps full, power off, maximum braking (POH Fig 5-11).

Landing note: if landing with flaps up, increase the approach speed by 9 KIAS and allow for 35 percent longer distances (POH Fig 5-11).

Ask the student:

- You must land with a flap failure. What does the POH say to do with speed and distance?

Source: POH Fig 5-5, POH Fig 5-7, POH Fig 5-11, PHAK 11-8

Slide 11: Airport Environment: Surface and Slope

ACS: AI.II.F.K2d

Performance charts typically assume paved, level, smooth, and dry runways. Any surface that is not hard and smooth increases the ground roll on takeoff (PHAK 11-12).

An upsloping runway gives a longer takeoff ground run but usually shortens the landing roll. A downslope does the opposite. Runway surface and slope are in the Chart Supplement (PHAK 11-13, PHAK Fig 11-17).

C172S notes: on a dry grass runway, increase takeoff distances by 15 percent of the ground roll figure (POH Fig 5-5) and landing distances by 45 percent of the ground roll figure (POH Fig 5-11).

Water on the runway reduces braking and can cause hydroplaning (PHAK 11-13).

Ask the student:

- The ground roll is 1,000 feet on pavement. About how long on dry grass for takeoff?

Source: PHAK 11-12, PHAK 11-13, POH Fig 5-5, POH Fig 5-11, PHAK Fig 11-17

Slide 12: Airport Environment: Wind

ACS: AI.II.F.K2d

A headwind lets the airplane reach lift-off speed at a lower groundspeed. A tailwind requires a higher groundspeed (PHAK 11-15).

A headwind that is 10 percent of the takeoff airspeed reduces takeoff distance about 19 percent. A tailwind that is 10 percent of takeoff airspeed increases it about 21 percent (PHAK 11-15, PHAK Fig 11-19).

C172S note: decrease distances 10 percent for each 9 knots of headwind. For tailwinds up to 10 knots, increase distances by 10 percent for each 2 knots (POH Fig 5-5).

Example: An 18 knot headwind: 18 divided by 9 = 2, times 10 percent = 20 percent shorter. A 4 knot tailwind: 4 divided by 2 = 2, times 10 percent = 20 percent longer.

Ask the student:

- Why is the tailwind correction so much larger than the headwind correction?

Source: PHAK 11-15, POH Fig 5-5, PHAK Fig 11-19

Slide 13: Loading and Weight

ACS: AI.II.F.K2e

Increased gross weight has a threefold effect on takeoff performance: higher lift-off speed, greater mass to accelerate, and increased retarding force from drag and ground friction (PHAK 11-15).

Excess weight also gives a longer takeoff run, reduced climb, shorter range, higher stalling speed, higher approach speed, and a longer landing roll (PHAK 10-2).

Example: C172S at sea level and 20 degrees C: total distance over a 50 foot obstacle is 1,690 feet at 2,550 lb (POH 5-14) and 1,205 feet at 2,200 lb (POH 5-16).

Ask the student:

- Name the three ways extra weight lengthens the takeoff.

Source: PHAK 11-15, PHAK 10-2, POH 5-14, POH 5-16

Slide 14: The Type Certificate Data Sheet

ACS: AI.II.F.K3, AI.II.F.R2

TCDS 3A12 page 1 says the data sheet "prescribes conditions and limitations under which the product for which the type certificate was issued meets the airworthiness requirements."

The C172S section (page 30) lists the Lycoming IO-360-L2A rated 180 horsepower, 100/100LL minimum grade fuel, 2,700 RPM engine limit, and airspeed limits of 105 KIAS maneuvering, 129 KIAS maximum structural cruising, 163 KIAS never exceed, and 85 KIAS flaps extended.

CG limits are published in the TCDS and in the POH (PHAK 10-3). Use the POH for the airplane you fly, and know the TCDS backs it up.

Ask the student:

- Where else, besides the POH, can you find the C172S CG limits?

Source: TCDS 3A12 p1, TCDS 3A12 p30, PHAK 10-3

Slide 15: Definitions: Datum and Arm

ACS: AI.II.F.K3

Read the PHAK definitions of datum (PHAK 10-5) and arm (PHAK 10-4) word for word. Point to the datum, the plus arm, and the minus arm in PHAK Fig 10-2.

Source: PHAK 10-4, PHAK 10-5, PHAK Fig 10-2

Slide 16: Datum and Arm in Plain Words

ACS: AI.II.F.K3

The datum is the zero point for all measurements. For the C172S it is the lower portion of the front face of the firewall (TCDS 3A12 p30).

The arm is how far an item is from the datum. Items behind the datum are plus, items in front are minus (PHAK 10-6).

Example: C172S arms from the TCDS: front seats 34.0 to 46.0 inches, rear seats 73.0 inches, fuel 48.0 inches aft of datum, and oil 13.1 inches forward of datum, which is a minus arm (TCDS 3A12 p30).

Analogy: hook a tape measure on the firewall and stretch it back. The number on the tape under each seat or tank is its arm.

Ask the student:

- Is the oil arm plus or minus in the C172S? Why?

Source: TCDS 3A12 p30, PHAK 10-6

Slide 17: Definitions: Moment and CG

ACS: AI.II.F.K3

Read the PHAK definitions of moment (PHAK 10-5) and CG (PHAK 10-4) word for word. The full CG definition goes on to say it is the mass center of the aircraft, it may be expressed in inches from the datum or in percent of MAC, and it is a three-dimensional point.

Source: PHAK 10-4, PHAK 10-5, PHAK Fig 10-3

Slide 18: Moment and CG in Plain Words

ACS: AI.II.F.K3

A moment is weight times arm. It measures how strongly a weight tries to rotate about a point (PHAK 10-6).

PHAK example: 50 lb placed 100 inches from the datum makes 5,000 in-lb. That same 5,000 in-lb can be balanced by 100 lb at 25 inches plus 50 lb at 50 inches (PHAK Fig 10-4).

To find the CG, divide total moment by total weight (PHAK 10-7).

Analogy: a playground seesaw. A light child far from the middle can balance a heavy child close to the middle. The seesaw balances where the moments are equal.

Ask the student:

- Which makes a bigger moment: 100 lb at 40 inches, or 50 lb at 95 inches?

Source: PHAK 10-6, PHAK 10-7, PHAK Fig 10-4

Slide 19: Definitions: CG Limits and Maximum Weight

ACS: AI.II.F.K3

Read the PHAK definitions of CG limits and maximum weight (PHAK 10-5). The ACS uses the words "maximum gross weight." The PHAK definition list uses "maximum weight," and its useful load definition refers to the "maximum allowable gross weight."

Source: PHAK 10-5, POH 2-8

Slide 20: C172S Weight and CG Limits

ACS: AI.II.F.K3, AI.II.F.R2

Normal category limits: maximum ramp weight 2,558 lb, maximum takeoff and landing weight 2,550 lb. Baggage area A (station 82 to 108) holds 120 lb, baggage area B (station 108 to 142) holds 50 lb, and the combined maximum is 120 lb (POH 2-8).

CG range: forward limit 35.0 inches aft of datum at 1,950 lb or less, with straight line variation to 41.0 inches at 2,550 lb. Aft limit 47.3 inches at all weights (POH 2-8, TCDS 3A12 p30).

Utility category has smaller limits: 2,200 lb maximum takeoff weight, aft limit 40.5 inches, and the baggage area must be empty with the rear seat unoccupied (POH 2-8, POH 2-9).

Ask the student:

- What is the forward CG limit at 2,550 lb?
- Why is the ramp weight 8 lb more than the takeoff weight?

Source: POH 2-8, TCDS 3A12 p30

Slide 21: Definitions: Basic Empty Weight and Useful Load

ACS: AI.II.F.K3

Read the PHAK definitions of basic empty weight (PHAK 10-4) and useful load (PHAK 10-5) word for word. The PHAK notes the useful load term applies to general aviation aircraft only.

Source: PHAK 10-4, PHAK 10-5, POH Fig 6-3

Slide 22: Empty Weight and Useful Load in Plain Words

ACS: AI.II.F.K3

Basic empty weight comes from the weight and balance records for your airplane as it is presently equipped. It includes unusable fuel and full oil (POH 6-10). TCDS Note 1 says the empty weight must include 18 lb of unusable fuel at 46.0 inches and 15.0 lb of full oil at 13.1 inches forward of datum (TCDS 3A12 p32).

Useful load is everything we can add: people, bags, and usable fuel.

Example: The POH sample airplane has a basic empty weight of 1,642 lb (POH 6-10). 2,550 lb maximum takeoff weight minus 1,642 lb = 908 lb useful load. The C172S holds 53 gallons usable fuel (TCDS 3A12 p30). At 6 lb per gallon that is 318 lb, leaving 590 lb for people and bags.

Analogy: a pickup truck with a payload rating. The truck itself is the empty weight. What you can put in the bed and the cab is the useful load.

Ask the student:

- Why must you never use the POH sample empty weight for your real flight? (PHAK 10-6 says sample problems are for guidance only.)

Source: POH 6-10, TCDS 3A12 p30, TCDS 3A12 p32, PHAK 10-6

Slide 23: Three Ways to Compute CG

ACS: AI.II.F.K4

Computational method: list each weight, multiply by its arm to get the moment, total the weights and moments, then divide total moment by total weight to get the CG (PHAK 10-7).

PHAK example: total weight 3,320 lb, total moment 281,430 in-lb. 281,430 divided by 3,320 = 84.8 inches, inside the 78 to 86 inch range (PHAK Fig 10-5).

Graph method: use the manufacturer's loading graph to find moments, then plot total weight and moment on the CG envelope (PHAK 10-7). Table method: the manufacturer gives tables of moments for each weight (PHAK 10-9).

Ask the student:

- What do you divide to get the CG?

Source: PHAK 10-7, PHAK 10-9, PHAK Fig 10-5

Slide 24: C172S Graph Method

ACS: AI.II.F.K4

Using the loading graph: find the weight, draw a line straight across until it meets the line for the item, then draw a line straight down to read the moment (PHAK 10-7).

The C172S loading graph has lines for pilot and front passenger, fuel, rear passengers, baggage area A, and baggage area B. The moment scale is load moment divided by 1,000 in pound-inches (POH Fig 6-4).

Ask the student:

- Two front seat occupants weigh 340 lb total. Walk me through finding their moment on the graph.

Source: PHAK 10-7, POH Fig 6-4

Slide 25: Aerodynamics: Forward CG

ACS: AI.II.F.K5

With forward loading, the tail must produce a greater down load. That adds to the wing loading, requires a higher angle of attack, adds drag, and raises the stall speed (PHAK 5-43).

The airplane stalls at a higher speed with a forward CG, and elevator control forces are higher (PHAK 5-44).

Beyond the forward limit the nose may be so heavy that it is difficult or impossible to flare for landing (PHAK 10-3).

Ask the student:

- Why does a forward CG raise the stall speed?

Source: PHAK 5-43, PHAK 5-44, PHAK 10-3

Slide 26: Aerodynamics: Aft CG

ACS: AI.II.F.K5

The airplane becomes less stable as the CG moves aft. If it moves far enough aft the airplane becomes unstable (PHAK 5-44).

Aft of the limit: reduced stability, extreme control difficulty, violent stall characteristics, and very light control forces that make it easy to overstress the airplane (PHAK 10-3).

Stall recovery gets harder as the CG moves aft, and a flat spin can develop that may be impossible to recover from (PHAK 5-43, PHAK 5-44).

An aft CG cruises faster because of reduced drag (PHAK 5-44). This is the trap: it feels better until you need control at low speed.

Ask the student:

- Which is more dangerous for stall recovery, forward or aft CG? Why?

Source: PHAK 5-43, PHAK 5-44, PHAK 10-3

Slide 27: Example: C172S Takeoff Setup

ACS: AI.II.F.S1, AI.II.F.K1

Use the sample problem from POH 5-3 and 5-4. Have the student find the 2,000 foot row and the 30 degree C column on Figure 5-5.

Given	Value
Takeoff weight	2,550 lb
Usable fuel	53 gal
Field pressure altitude	1,500 ft
Temperature	28 C (16 C above standard)
Wind	12 kt headwind
Field length	3,500 ft

Ask the student:

- Why do we read 2,000 feet and 30 degrees instead of 1,500 feet and 28 degrees?

Source: POH 5-3, POH 5-4, POH Fig 5-5

Slide 28: Example: Takeoff with Wind Correction

ACS: AI.II.F.S1

From Figure 5-5 at 2,000 feet and 30 degrees C: ground roll 1,285 feet, total to clear a 50 foot obstacle 2,190 feet (POH 5-4).

Wind correction from Note 3: 12 knots divided by 9 knots, times 10 percent, is about a 13 percent decrease (POH 5-4).

Corrected ground roll: 1,285 minus 167 = 1,118 feet. Corrected total distance: 2,190 minus 285 = 1,905 feet (POH 5-5). Both fit in the 3,500 foot field.

Ask the student:

- Would you plan to use the headwind credit if the wind was gusty and variable? Why or why not?

Source: POH 5-4, POH 5-5, POH Fig 5-5

Slide 29: Example: Fuel and Distance to Climb

ACS: AI.II.F.S1

Figure 5-7 shows a climb from 2,000 to 8,000 feet takes 2.2 gallons and 15 nautical miles at standard temperature (POH 5-6).

The chart note says increase time, fuel, and distance by 10 percent for each 10 degrees C above standard. At 16 degrees above standard that is a 16 percent increase (POH 5-6, POH Fig 5-7).

Fuel: 2.2 plus 0.4 = 2.6 gallons. Distance: 15 plus 2.4 = 17.4, rounded up to 18 nautical miles (POH 5-6).

Example: Show how to subtract: at 8,000 feet the chart reads 3.0 gallons and 19 nm from sea level. At 2,000 feet it reads 0.8 gallons and 4 nm. The POH sample uses 2.2 gallons and 15 nm for the climb between them (POH Fig 5-7).

Ask the student:

- Why do we subtract the departure field line from the cruise altitude line?

Source: POH 5-6, POH Fig 5-7

Slide 30: Example: Cruise Performance

ACS: AI.II.F.S1

Cruise is planned at 7,500 feet pressure altitude, 16 degrees C. The cruise chart is entered at 8,000 feet and 20 degrees C above standard, which most nearly match (POH 5-5).

At 2,600 RPM: 64 percent power, 117 knots true airspeed, 8.9 gallons per hour (POH 5-5, POH Fig 5-8).

With a 10 knot headwind, groundspeed is 117 minus 10 = 107 knots (POH 5-7).

The range profile shows lower power settings save fuel and give longer range (POH 5-5).

Ask the student:

- If you reduce power to save fuel, what happens to true airspeed and time en route?

Source: POH 5-5, POH 5-7, POH Fig 5-8

Slide 31: Example: Total Fuel Required

ACS: AI.II.F.S1

Cruise distance: 360 minus 18 climb miles = 342 nautical miles (POH 5-6).

Cruise time: 342 nm divided by 107 knots = 3.2 hours. Cruise fuel: 3.2 hours times 8.9 GPH = 28.5 gallons (POH 5-7).

Reserve: 45 minutes divided by 60, times 8.9 GPH = 6.7 gallons (POH 5-7).

Total: 1.4 + 2.6 + 28.5 + 6.7 = 39.2 gallons, less than the 53 gallons usable (POH 5-7).

The POH says ground speed checks in flight give a better basis for estimating time and fuel (POH 5-7). This table is the fuel section of the nav log the student will build in Task II.I.

Leg	Math	Gallons

Start, taxi, takeoff	POH allowance	1.4
Climb	2.2 + 16%	2.6
Cruise	342 nm / 107 kt = 3.2 hr x 8.9 GPH	28.5
Reserve	45/60 x 8.9 GPH	6.7
Total		39.2

Ask the student:

- Where does the 1.4 gallons for start, taxi, and takeoff come from? (POH Fig 5-7 note)

Source: POH 5-6, POH 5-7, POH Fig 5-9

Slide 32: Example: Landing Distance

ACS: AI.II.F.S1

Destination: 2,000 feet pressure altitude, 25 degrees C, 3,000 foot field. The POH reads the chart at 2,000 feet and 30 degrees C: ground roll 650 feet, total over a 50 foot obstacle 1,455 feet (POH 5-8).

A wind correction can be made with Note 2 the same way as for takeoff (POH 5-8).

Ask the student:

- The runway is dry grass. What does Figure 5-11 say to add?

Source: POH 5-8, POH Fig 5-11

Slide 33: Example: C172S Sample Loading

ACS: AI.II.F.S2

Walk through POH Figure 6-3 line by line. Basic empty weight includes unusable fuel and full oil. Fuel is figured at 6 lb per gallon (POH 6-10).

Ramp weight is 2,558 lb with a moment of 113.2. Subtract 8 lb and 0.4 moment for engine start, taxi, and runup. Takeoff weight is 2,550 lb with a moment of 112.8 (POH 6-10).

Ask the student:

- Why is the ramp weight allowed to be 2,558 lb?

Source: POH 6-10, POH Fig 6-3

Slide 34: Example: Plot on the Envelope

ACS: AI.II.F.S2

Locate 2,550 lb and 112.8 on the Center-of-Gravity Moment Envelope. The point falls within the envelope, so the loading is acceptable (POH 6-10, POH Fig 6-7).

To get the CG in inches, multiply the moment back by 1,000 and divide by weight: 112,800 divided by 2,550 = 44.2 inches. At 2,550 lb the limits are 41.0 to 47.3 inches (POH 2-8), so 44.2 is inside.

Ask the student:

- What would it mean if the point fell to the right of the envelope?

Source: POH 6-10, POH Fig 6-7, POH 2-8

Slide 35: Check Every Phase: Fuel Burn

ACS: AI.II.F.S2, AI.II.F.R6

The ACS requires checking weight and balance for all phases of flight. In flight, fuel burn is normally the only weight change (PHAK 10-2).

C172S fuel is in two wing tanks at 48.0 inches aft of datum (TCDS 3A12 p30). Most small aircraft have tanks close to the CG, so burning fuel does not move the CG much (PHAK 10-11).

Example: Worked example using the POH sample load: burn 15 gallons (90 lb). Moment removed: 90 x 48.0 = 4,320 in-lb, or 4.3 on the /1000 scale. Landing: 2,460 lb and 108.5. CG: 108,500 divided by 2,460 = 44.1 inches. The forward limit at 2,460 lb is 35.0 plus (510 / 600 x 6.0) = 40.1 inches and the aft limit is 47.3 inches (POH 2-8), so the landing CG is in limits.

Ask the student:

- Fuel burns from tanks at 48.0 inches and the CG is 44.2. Does the CG move forward or aft as fuel burns? (Fuel is aft of the CG, so the CG moves slightly forward.)

Source: TCDS 3A12 p30, PHAK 10-11, POH 2-8, POH Fig 6-4

Slide 36: Fixing an Out-of-Limits CG

ACS: AI.II.F.S2, AI.II.F.R6

Weight shift formula: weight shifted divided by total weight equals change of CG divided by distance the weight is shifted (PHAK 10-11).

PHAK example: total weight 7,800 lb, CG 81.5 inches, aft limit 80.5 inches. Cargo moves from station 150 to station 30, a 120 inch shift. Weight to be shifted / 7,800 = 1.0 / 120, so weight to be shifted = 65 lb (PHAK 10-11).

Adding or removing weight uses: added weight / new total weight = change of CG / distance between the weight and the old CG. If the CG moves aft, add the change. If it moves forward, subtract it (PHAK 10-11).

In the C172S, the practical fixes are moving bags forward, swapping heavier people to the front seats, or reducing weight (PHAK 10-3).

Ask the student:

- The CG is too far aft. Name two ways to fix it in a C172S.

Source: PHAK 10-10, PHAK 10-11

Slide 37: Risk: Charts vs. Real Life

ACS: AI.II.F.R1, AI.II.F.R3

R1: Wrong chart, wrong row, or skipped notes give a wrong answer. Always read conditions and notes and use conservative values (POH 5-4).

R3: The POH says variables such as leaning technique, engine and propeller condition, and turbulence can cause variations of 10 percent or more in range and endurance, so plan conservatively (POH 5-3). Charts come from a new airplane flown with average skill (PHAK 11-19).

Mitigation: add a personal margin, plan an abort point on the runway, and decide before takeoff what you will do if you are not at the expected speed.

Example: Scenario for discussion: 32 degrees C, field elevation 3,000 feet, 2,400 foot grass strip, airplane near maximum weight. Have the student find the chart number, apply the grass note, then decide go or no-go and what margin they want.

Ask the student:

- The chart says you need 2,100 feet and the runway is 2,400 feet. Would you go? What would change your mind?

Source: POH 5-3, POH 5-4, PHAK 11-19

Slide 38: Risk: Limits and Overweight

ACS: AI.II.F.R2, AI.II.F.R4

R2: 14 CFR 91.9 requires the pilot in command to comply with the operating limits in the approved AFM, and these include weight and balance (PHAK 10-4).

Airspeed limits are limits too. Note the C172S maneuvering speed drops as weight drops: 105 KIAS at 2,550 lb, 98 KIAS at 2,200 lb, and 90 KIAS at 1,900 lb (POH 2-4).

R4: Operating above maximum weight compromises structural integrity and hurts performance (PHAK 10-1). An overloaded airplane may not leave the ground, or may show unusually poor flight characteristics (PHAK 10-2).

Example: Scenario for discussion: four adults and bags for a weekend trip, full tanks. Have the student compute it with the POH sample empty weight, find the overage, and choose a fix (less fuel and a fuel stop, fewer bags, or fewer people).

Ask the student:

- If you must take off fuel to stay under weight, what new risk do you create, and how do you manage it?

Source: PHAK 10-1, PHAK 10-2, PHAK 10-4, POH 2-4

Slide 39: Risk: CG Limits and Shifting Weight

ACS: AI.II.F.R5, AI.II.F.R6

R5: Operation with the CG outside the approved limits results in control difficulty (PHAK 10-1). Aft CG can cause violent stall characteristics and very light control forces (PHAK 10-3).

R6: The CG depends on how weight is distributed. As load is shifted or used, the CG moves (PHAK 10-3). Adding or removing weight changes both weight and CG (PHAK 10-11).

Baggage area limits: area A holds 120 lb and area B holds 50 lb, with 120 lb combined maximum (POH 2-8). Exceeding these overloads the floor even if the total weight is fine.

Example: Scenario for discussion: after you finish the weight and balance, a passenger adds a 40 lb cooler to baggage area B. Ask what changed, whether the area B limit is exceeded with what is already there, and how to recompute.

Ask the student:

- After loading is finished, a passenger adds a bag. What do you do?

Source: PHAK 10-1, PHAK 10-3, PHAK 10-11, POH 2-8, PHAK Fig 10-1

Slide 40: Common Errors

ACS: AI.II.F.S1, AI.II.F.S2

Using the POH sample empty weight. Fix: use the airplane's current weight and balance record. Sample problems are guidance only (PHAK 10-6).

Skipping the notes (wind, grass, flaps up). Fix: read every note before using a number (POH Fig 5-5).

Not correcting climb numbers for temperature. Fix: add 10 percent per 10 degrees C above standard (POH 5-6).

Forgetting start and taxi fuel. Fix: subtract 8 lb at ramp for weight and balance (POH 6-10) and add 1.4 gallons for fuel planning (POH Fig 5-7).

Source: PHAK 10-6, POH 5-4, POH 5-6, POH 6-10, POH Fig 5-5, POH Fig 5-7

Slide 41: Review Questions

ACS: AI.II.F

Answers: (1) higher temperature, higher altitude, and higher humidity (PHAK 11-2, PHAK 11-5). (2) Lower portion of the front face of the firewall (TCDS 3A12 p30). (3) Total moment divided by total weight (PHAK 10-7). (4) 18 divided by 9 times 10 percent = 20 percent shorter (POH Fig 5-5).

Source: PHAK 11-2, PHAK 10-7, TCDS 3A12 p30, POH Fig 5-5

Slide 42: References

ACS: AI.II.F

Review the sources. Remind the student that the C172S pages are for training and that the POH and weight and balance records for the actual airplane always govern.

Source: See lesson plan for full reference list

7. Instructor Actions

- Present each slide in order and use the speaker notes.
- Demonstrate one chart reading and one weight and balance problem on the whiteboard.
- Use the model airplane and ruler to show datum, arm, and moment.
- Ask the review questions on each slide and correct errors right away.
- Give the student a new scenario and watch them solve it without help.
- Assess using the completion standards and debrief.

8. Student Actions

- Follow along in the POH and find each chart.
- Take notes and ask questions.
- Work the takeoff, climb, cruise, fuel, and landing problems.
- Complete a weight and balance for the real airplane with real weights.
- Explain the risks and mitigations in their own words.

9. Common Errors

Using the POH sample empty weight. Use the current weight and balance record for the airplane. Sample problems are for guidance only (PHAK 10-6).

Ignoring chart conditions and notes. Read the conditions block and every note before reading a number (POH Fig 5-5, POH Fig 5-11).

Reading the wrong row or column. Use the next higher weight, altitude, and temperature, or interpolate carefully (POH 5-4, PHAK 11-20).

No temperature correction on climb. Add 10 percent per 10 degrees C above standard (POH Fig 5-7).

Forgetting start and taxi fuel. Subtract 8 lb at ramp (POH 6-10) and plan 1.4 gallons (POH Fig 5-7).

Checking CG only at takeoff. Check the landing weight and CG too, after fuel burn.

Exceeding the baggage limits. Area A 120 lb, area B 50 lb, 120 lb combined (POH 2-8).

10. Conclusion and Review

Summarize: performance data comes from the POH and assumes a new airplane, average technique, and the chart conditions. Density altitude, technique, configuration, runway, wind, and weight all change the numbers. Weight and balance uses weight times arm equals moment, and CG equals total moment divided by total weight. Limits come from the POH and TCDS and they are mandatory.

Review using slide 41. Then give the student a new scenario (different weights, temperature, and runway) and have them work it from start to finish. Preview Task II.I, where these numbers go into a full nav log.

11. Completion Standards

- The student explains how to use the charts, tables, and data in the POH (AI.II.F.K1).
- The student explains how atmosphere, technique, configuration, airport environment, and loading affect performance (AI.II.F.K2).
- The student defines and explains basic empty weight, maximum gross weight, arm, moment, datum, CG and CG limits, and useful load (AI.II.F.K3).
- The student computes CG by computation and graph methods (AI.II.F.K4) and explains the aerodynamic effects of CG location (AI.II.F.K5).
- The student identifies, assesses, and mitigates the risks in AI.II.F.R1 through R6.
- The student correctly uses the C172S charts with no more than minor reading errors that they catch and fix (AI.II.F.S1).
- The student computes weight and balance, corrects an out-of-limits CG, and confirms limits for all phases of flight (AI.II.F.S2).

12. References

- FAA-S-ACS-25, Flight Instructor for Airplane Category Airman Certification Standards (November 2023), Area II, Task F.

- FAA-H-8083-25C, Pilot's Handbook of Aeronautical Knowledge (PHAK): pages 5-43, 5-44, 10-1 to 10-11, 11-1 to 11-20. The October 2025 PHAK addendum does not change these chapters.
- FAA-H-8083-9, Aviation Instructor's Handbook (AIH): pages 5-9 (lesson introduction) and 7-9 to 7-10 (lesson plan format).
- Cessna Model 172S NAV III (G1000) Pilot's Operating Handbook pages, P/N 172SPHAUS, as printed in the Cessna 172S NAV III Information Manual 172SIMAUS, Revision 3: Section 2 (2-4, 2-8), Section 5 (5-1 to 5-8, Figures 5-5, 5-7, 5-8, 5-9, 5-11), Section 6 (6-10, 6-12, 6-15, Figures 6-3, 6-4, 6-7). For training only. Always use the POH/AFM for the airplane you fly.
- FAA Type Certificate Data Sheet 3A12, Revision 89 (September 15, 2026): page 1 and Model 172S data on pages 30 and 32.